Emergency AI co-scientist — overview and plan for next six weeks

Summary

The system is an **AI co-scientist that decides what to pursue and what to measure next** under outbreak conditions: incomplete knowledge of the target, evidence that conflicts and arrives late, a hard deadline, a fixed budget of actions, and one shot at the laboratory. The central object is not a prediction but an allocation.

Four things changed in the last week, and they set the plan.

The acquisition machinery now works on the right molecule class. On two macrocyclic libraries the policies reach 0.64–0.71 recall against a random baseline near 0.27 at matched budget, significant across pools. This is the first result in the project on macrocyclic substrates, and on one of them the features are named non-canonical monomers rather than amino-acid letters. It did not require SynX.

The decision arm's main readout turned out to be unmeasurable. Forbidding evidence retrieval entirely, the co-scientist still named the reference-best target in 16 of 16 episodes having bought nothing, while the evidence-using arm bought 90 items across 15 episodes for the identical score. The difference is 0.0. Target-selection agreement is saturated by what the model already knows, so it cannot detect evidence-driven decision quality in either direction. **Building a readout that is not saturated is the single most important task of week 5.**

Three claims were withdrawn, with the arithmetic shown. They are listed in §4.2. Two were mine and one was a formula error that had been reported in every summary.

The engineering is now in a state where results regenerate rather than accumulate. The full grid is 2,556 graded cells, reproducible in four minutes on one laptop, every artifact stamped with its host and commit, 39 tests on the invariants that would invalidate a result.

Three external dependencies remain overdue and none has an internal workaround. §6 states each and what it blocks.

1. What the system is

1.1 The decision it makes

Each episode, under an eight-round deadline and a fixed action budget with the remaining budget visible in its state, the co-scientist chooses one of six actions:

ACTION	WHAT IT DOES
`retrieve_evidence`	Buy sourced evidence for one named target on one axis
`request_structure`	Request a structural or conformational calculation
`rerank_targets`	Revisit the ranking against evidence observed so far
`invoke_design_tool`	Propose a focused library against the leading target
`recommend_experiment`	Recommend the single round, and state what it can substantiate
`stop`	Stop and report the dossier with its open uncertainties

The action budget binds independently of the deadline, and whichever binds first ends the episode. Until this week the budget could never bind — with eight rounds, eight actions and one action per round, the exhaustion branch first held at round nine, which the loop never reached. The budget the prompt described to the policy was fiction; it now binds and is tested.

1.2 What it reasons over

A **seven-axis dossier per candidate target**. The axes are the Week 4 selection rubric, and each is in the state because it gates a decision rather than because it is interesting:

AXIS	WHAT IT GATES
`reagent_tractability`	**Hard gate.** An untractable reagent ends the campaign regardless of the biology.
`essentiality`	A binder to a dispensable protein cannot become a therapeutic.
`conservation`	Gates every family-wide and prototype-pathogen claim.
`accessibility`	Sets whether a permeability story is required.
`assay_availability`	Without one, a single round can only substantiate enrichment.
`public_labels`	Determines whether transfer tests are runnable now or blocked on SynX.
`precedent`	Supplies positive controls and de-risks interpretation of a failure.

Each claim on an axis carries a source, an availability date, one of four strength levels (`unknown`, `weak`, `moderate`, `strong`), a `verified` flag, and an optional pointer to the claim it contradicts. Only `reagent_tractability` is a hard gate: fail it and the campaign ends however strong the other six look.

The rubric is also the reason the saturation problem in §4.2 has a tractable fix. Which axis is currently gating a given decision depends on the evidence state the co-scientist has been handed, not on the target's identity — so it is a question pretraining cannot answer, which is what §5.1 builds on.

1.3 Three rules enforced in code, not by convention

- **Date-limited retrieval.** The dossier refuses evidence dated after the decision under consideration, and records the refusal rather than discarding it. The refusal count is deliberately *not* shown to the policy: telling it that three claims exist which it may not see is information from after the decision clock arriving through a side channel.
- **Contradictions are retained and consequential.** Conflicting claims coexist rather than overwrite. As of this week a contested axis no longer counts as substantiated — previously the ranking read a contested axis as settled while the state recorded that it was not, which made retention decorative.
- **An uncited claim cannot satisfy a gate.** Claims asserted without a citation are carried with `verified=false` and excluded from substantiation.

Each rule is pinned by a test that would fail if the rule were removed.

2. How it is graded without a laboratory or a reviewer

Knowing whether a scientific decision was a good one normally costs either a bench experiment or an expert's time. Both are slow and expensive, and expert judgement has inter-rater problems that are themselves a project risk.

The alternative used here: take a dataset in which someone has *already measured* the outcome for every candidate, hide all of it behind an evaluator, and let the co-scientist pay to reveal a few at a time under a budget. Every choice it makes is then scored exactly against the outcomes that were withheld. Grading is deterministic, needs no bench and no reviewer, and can be repeated as often as the question requires.

A note on one word, because it is overloaded. "**Policy" below means the decision rule — the thing choosing the next action**, in the machine-learning sense. It has nothing to do with public-health policy or with WHO guidance; the project's one WHO reference is the prototype-pathogen rationale recorded in the evidence packets, which is a justification for family choice and is unrelated to this section.

Four substrate families run through one replay engine: the original pilot landscape, the pathogen landscapes, the method-development landscapes carrying measured binding endpoints, and the macrocyclic display libraries added this week.

Two falsification controls were added this week, and both are load-bearing.

- **Permuted outcomes.** Shuffle the outcome vector before the episode and grade against the shuffle. A policy that learns must collapse to the random baseline. Measured over twenty episode coordinates: 0.804 on the real signal, 0.265 permuted, 0.263 random — the control lands 0.002 from random, which is what it is for. A policy that did *not* collapse would be scoring on index order or tie-breaking rather than on the evidence it paid for.
- **Zero retrieval.** Switch off the co-scientist's ability to look anything up, hand it an empty dossier, and require a decision anyway. In plain terms it is a cheating detector: if the system still names the target the full evidence base supports, it did not reason its way there, because it bought nothing to reason from — it already knew the answer from pretraining. It fired, in 16 of 16 episodes. This is the audit the architecture document named as necessary and left undone, and it is the reason §5.1 leads the plan.

Controls write to their own files and are excluded from every aggregate, because a falsification check is not a result.

3. Architecture

One replay engine plus a registry, with the decision layer kept entirely separate.

COMPONENT	ROLE
`arms.py`	The four substrate families as frozen records. Their differences are data — paths, outcome column, caveats, config hashes — not code. No I/O, no numerical dependencies.
`binding.py`	The replay engine, and one `run_arm` driven by that registry. The four named entry points are one-line delegations.
`streams.py`	One independent random stream per `(seed, purpose, index)` coordinate.
`data.py`	Staging, funnelled through a single exit that digests artifacts after writing them.
`sweep.py`	Grid enumeration and parallel execution, one code path for every arm.
`dossier.py`, `targets.py`, `emergency.py`	The decision layer. No numerical dependencies in the state object; it was written before the policy that consumes it.
`common.py`	The only inference client. Provider failures retry; a policy failure never does.
`summarize.py`, `calibration.py`	Evaluation, kept off the policy path.

The consolidation happened this week because the copies had already drifted: the sweep knew three arms, the summariser five names, and only one module knew that one arm's outcome column is named differently. Equivalence of the new single path against each of the four predecessors was verified field-for-field across twelve cells before the old bodies were deleted.

Reproducibility. Every run records host, platform, commit, Slurm identifiers, and whether the working tree was modified — the last being the one to read first, since a run from a modified tree is not reproducible from the commit it names. 2,557 artifacts currently carry a clean commit.

4. What has been measured

4.1 What survives

The acquisition machinery transfers to macrocyclic substrates. At 128 labels against a random baseline near 0.27 at the same budget:

LIBRARY	POOLS	GREEDY	UCB	DIVERSE	SIGNIFICANCE
Selection-enrichment library	20	0.640	0.657	0.027	p = 0.0001
Activity library, non-canonical monomer features	15	0.688	0.711	0.260	p = 0.0012, 0.0008

This is the first evidence in the project that the machinery works on the molecule class the project is about, and the second row is the first result on non-canonical features.

A sharp negative in the same table. Pure-distance acquisition is not merely unhelpful on the enrichment library — it is harmful: 0.027 against random's 0.267, worse in 20 of 20 pools, p = 0.0001 in the wrong direction. It spends the whole budget away from the hits. On the activity library it is indistinguishable from random instead.

The eight-label row is recorded before any policy acts, and is not about the surrogate. This section previously read "the surrogate has no discriminative power at the smallest budget", on the grounds that every policy returns identical recall at eight labels in 21 of 105 rows of the gain table. The identity is real; the attribution was wrong. `replay` calls `snapshot()` before the acquisition loop opens, so the eight-label record is the shared random initialisation — `round: 0`, while the first acquisition is `round: 1` — taken before any `pick_batch` call has executed. Every policy agrees there by construction at every seed, which is the property `test_all_policies_still_open_from_identical_observations` exists to pin, and it says nothing whatever about the surrogate's power.

The genuine smallest decision budget is **sixteen** labels: one batch of eight chosen by a policy, with the surrogate fitted on eight. Measured there, the largest paired gain over random on any landscape is **+0.0225**; signal emerges at thirty-two and is substantial by sixty-four. The clause that survives is the last one — those rows carry no policy information — and it is now pinned by a test rather than asserted.

This is the fourth claim of the same shape to be withdrawn here: a quantity a feedback-blind policy reproduces by arithmetic, reported as a measurement. The others are in §4.2.

4.2 Three claims withdrawn

The 1.1–1.4× selected-versus-returned gap is an accounting identity, not a measurement. The returned set is always a subset of the selected set and recall divides both by the same pool-wide hit count, so the ratio is monotone in set inclusion and never falls below 1 in any of 368 recorded runs. The feedback-

blind random policy, whose selections are identical at every delay, reproduces the entire band: 1.297 against 1.312 for the informed policies. A quantity a feedback-blind policy can produce is not a measurement of how feedback delay degrades a decision.

The matched effect of delayed feedback is null. Holding budget and metric fixed: mean paired difference -0.011 over 534 pool comparisons, delay helping in 13 cells and hurting in 18, Wilcoxon $p = 0.31$. The random arm's paired difference is exactly 0.0, which is what makes it usable as the null band.

The attainable-headroom formula was wrong, and the metric correction is not novel. The reported bound was the inverse labelled fraction unconditionally; the attainable bound is the tighter of that and the inverse hit threshold. The two agree at the terminal checkpoint of every configured landscape, which is why no published figure changes, and diverge at the early checkpoints where raw multiples look most impressive — at eight labels in a 512 pool it claimed 64× of headroom where 20× exists. Separately, the chance-and-ceiling normalisation we had treated as a contribution is published prior art from active-learning benchmarking and from the enrichment-metrology literature. What remains ours is applying it per checkpoint and showing it reorders *published* results.

4.3 What has not been measured at all

- ~~The co-scientist's policy has never run on a local model on GPU.~~ **Now measured, and the answer is negative.** The pinned Qwen3-4B serves on an L40S and the policy path completes end to end — 40 probes, 40 calls, no transport retries, nothing content-filtered, artifact recorded — for twenty-three cents. The policy is far worse than the baselines it was meant to exceed: challenger-stratum attainment **−0.938** on Henipavirus against a leader-first ceiling of -0.363, where the fixed rubric reaches -0.229 and the lattice policy $+0.365$. Below the bound a leader-only policy is held to, and below the zero that marks the average purchase on offer.

The failure is state-tracking, not reasoning depth. It chose `reagent_tractability` in **39 of 40** probes — the first axis in declaration order, which is also the hard gate — so it holds one heuristic and applies it unconditionally. Of 19 purchases that moved nothing, **9 were requests for a claim already observed**, while the state handed to it carries `retrievable_axes` listing precisely what remains. The other 10 were offered and moved nothing.

Two consequences. The route to the volumes a trained policy needs does not run through this model. And a policy keying on the first-listed axis is keying on *presentation*, while the nuisance group permutes the dossier order and not the axis order — a real gap in the group, and exactly where this model is vulnerable.

Getting there required correcting the pinned environment, which had never been executed: `requirements.llm.lock.txt` resolves to transformers 5.19.0 against a vLLM 0.10.2 that calls an attribute transformers 5 removed, and stepping back cascades through tokenizers and huggingface-hub for four layers. The working resolution is transformers 4.57.6, tokenizers 0.22.2, huggingface_hub 0.36.2 — the last being two major versions from what the lock carries. The same failure is waiting for TRAIN-10 on the cluster, which builds `.venv-llm` from that file.

- **No macrocycle from the SynX platform** has been represented or measured. Every platform-supplied chemistry field remains null pending the campaign export.
- The **physics-based non-canonical track** is untouched.
- The structural arm is retired as a scientific component.

5. The six-week plan

5.1 The problem that sets the order

The decision arm's readout is saturated. Until there is a readout the zero-retrieval control does not saturate, no number from the decision arm can be attributed to evidence, and the project's primary research question has no instrument. Everything in week 5 serves that, and nothing downstream is worth building first.

Three candidate readouts, in descending order of how much they ask:

1. **Which axis it chooses to resolve next**, scored against which axis actually gates the decision. The co-scientist cannot know this from pretraining, because it depends on the evidence state it has been handed, not on the target's identity.
2. **Decision quality under corrupted and conflicting evidence.** Pretrained knowledge is a liability rather than an advantage when the supplied evidence contradicts it; this turns the saturation into a measurable failure mode.
3. **Anonymised targets.** Replace names with opaque identifiers and re-run. Compares directly against the named condition and isolates how much of the performance is the name.

5.2 Week by week

W5, 26–30 October — make the decision arm measurable.

ID	TASK	ACCEPTANCE
COSCI-20	Next-axis readout: score the chosen axis against the gating axis	Zero-retrieval control scores at or near chance; evidence arm beats it with a stated interval
COSCI-21	Anonymised-target condition	Named versus anonymised run on the same packets, difference reported with its interval
COSCI-22	Corrupted-evidence condition: supplied evidence contradicts the pretrained prior	Appropriate-update and hold rates separate the policies; no metric reads 1.0 for all of them
COSCI-23	Procedural episode generator: vary packet, budget, delay, initial evidence, corruption	≥200 graded episodes from the existing packets, with the generation parameters recorded per episode
FCAST-10	Re-score one published table under the corrected attainment metric	A published ranking either moves or demonstrably does not; result written up either way

Dependency: none external. This week is fully unblocked.

W6, 2–6 November — the local-model policy, and the trained arm.

ID	TASK	ACCEPTANCE
TRAIN-10	Stand the pinned local model up on GPU and validate the policy path end to end	One non-smoke agent episode per arm, artifact-recorded, from the local model
TRAIN-11	Cost-matched comparison: policy versus deterministic baselines at equal spend	Comparison reported in actions and tokens, not wall-clock
TRAIN-12	Supervised fine-tune on filtered successful episodes as the mandatory baseline	Held-out episode score against the untrained policy, ≥5 seeds
COSCI-24	Full decision-arm sweep with the generator from COSCI-23	Paired statistics with the unit of analysis stated

Dependency: **confirmed trainable-model access with an approved spending ceiling.** Modal is authenticated and is the fallback lane if cluster GPU does not materialise.

W7, 9–13 November — London week. Reduced compute capacity; lab-facing.

ID	TASK	ACCEPTANCE
LAB-*	SynX engagement	Named experimental owner and a booked slot, or a written re-specification
MOL-01..04	Platform chemistry representation, if the export has arrived	Building-block registry populated; otherwise the deliverable is re-specified against public data at the W7 gate
GATE-*	Mid-point gate: confirm or re-specify the platform-improvement deliverable	A decision recorded either way

Treat compute-heavy work as unavailable this week and schedule accordingly.

W8, 16–20 November — consolidate and stress.

ID	TASK	ACCEPTANCE
COSCI-25	Historical reconstruction episode on evidence restricted to an early window	Decision trace and its leakage audit, with the no-retrieval floor reported beside it
MOL-10	Transfer test: canonical-trained surrogate applied to the non-canonical library	Reported with the geometry bound; a null stated as a null
SAFE-*	Safety benchmark preparation	Preparation artifacts committed, with scope and limits stated
FCAST-11	Power analysis and the unit-of-analysis statement for every headline comparison	Every reported comparison carries an interval and a stated independent unit

W9, 23–27 November — writing, with results frozen.

ID	TASK	ACCEPTANCE
PAPER-01	Draft the contribution: environments, the two controls, the metric audit	Complete draft, every claim naming the file and run that produced it
PAPER-02	Reproducibility appendix: regenerate every figure from a clean checkout	A clean-clone regeneration that matches, or the differences enumerated
PAPER-03	Prior-art section, written against sources actually read	No citation in the draft that has not been fetched and checked

Freeze results at the start of this week. Anything not measured by 21 November does not go in.

W10, 30 November–4 December — handoff.

ID	TASK	ACCEPTANCE
PAPER-04	Presentation and final write-up	Delivered
GATE-99	Final reproducibility check and artifact handoff	Repository, artifacts and documents handed over with provenance intact

5.3 What is deliberately not in the plan

- Any multi-agent decomposition. The episode is a tightly coupled sequential decision over one evidence state, which is the regime where the current evidence favours a single policy with good tools; decomposition would be a cost multiplier.
- Any new benchmark-breadth claim. Our pools are sub-samples of single measurement tables, not independent replicates, so breadth is not ours to claim.
- The physics-based non-canonical track, unless the export arrives by W7.

6. Dependencies, risks, and what I need decided

#	DEPENDENCY	BLOCKS	CONSEQUENCE IF IT DOES NOT ARRIVE
1	**Campaign export**	MOL-01..04, platform-improvement deliverable	The deliverable is re-specified against public data at the W7 gate. A minimal subset this week beats a complete export in three weeks.
2	**Trainable-model access with a spending ceiling**	TRAIN-10..12	The trained-policy arm does not happen; the contribution narrows to environments plus the metric audit. Modal is the fallback.
3	**Named SynX experimental owner and a booked slot**	LAB-*, the prospective check	No prospective validation; conclusions stay retrospective and must say so.

Three decisions requested at Thursday's gate.

1. Which of the three candidate readouts in §5.1 to treat as primary. My recommendation is the next-axis readout, because it is the one the co-scientist cannot answer from the target's name.
2. Whether to re-specify the platform-improvement deliverable now against public data rather than hold W7 open for the export.
3. Whether the trained-policy arm is in scope. If access is not confirmed this week, I would rather cut it deliberately than discover in W8 that it was never going to run.

7. Capacity, stated plainly

Six weeks, one person, with W7 substantially consumed by travel. The plan above is roughly five productive weeks of work. The W5 block is the part I would protect at the expense of everything else: without a readout the control does not saturate, the decision arm produces numbers that cannot be attributed to evidence, and that is the project's primary question.

The macrocyclic result and the two falsification controls are, as of this week, the contribution most likely to survive review. They are also the parts that required nothing from SynX, which is worth weighing when deciding how much of the remaining six weeks to spend waiting.